

H-60 Phase 0 Fixes

Plain-English version · August 16, 2026 · Companion to *H-60 Improvement Plan.md* (Phase 0 section has the file-and-line detail)

What Phase 0 is: a batch of small, surgical bug fixes — no redesigns, no restructuring, no gameplay changes beyond "the broken thing now works." Every item below is a confirmed bug found by reading the code or reported on GitHub. Each gets its own commit so anything can be picked out or reverted individually. Two former Phase 0 items — the FLIR vehicle tracking (#538) and the missing copilot console (#556) — are already fixed and sitting in PR #559, so they're out of this batch.

The fixes, grouped by system

RESCUE HOIST · 5 BUGS

The hoist misbehaves for anyone who isn't the pilot, and on dedicated servers

BROKEN NOW: The "raise hook" control accidentally runs the *lower* code when another player's machine owns the helicopter — a copy-paste slip. The scripts also ignore which helicopter they were told to work on and grab "the vehicle the player is sitting in" instead, which on a dedicated server is nobody, so the hoist can dead-stop. The winch loop runs as fast as the computer allows (wasted performance, and winch speed varies by machine). The hoist menu entry shows hardcoded English even though a translated name is prepared one line above it. And the same hoist actions are registered twice through two different systems, so they can appear doubled.

THE CHANGE: raise raises, the scripts use the helicopter they're given, the winch runs on a proper timer, the translated name is used, and the duplicate registration goes away.

YOU'LL TEST: raise/lower from a crew seat on a dedicated server with another player (or a second machine) — the exact setup where it breaks today.

Startup code leaks and cross-contaminates between helicopters

BROKEN NOW: Engine 2's power state is never set up (engine 1's is set up twice — another copy-paste slip). Every time anyone boards, an extra input-event listener is added and never removed, so they pile up over a mission — this matches the old report (#442) of network load climbing "when the pilot enters." And several per-helicopter values (which event listener belongs to it, last fuel level, last altitude) are stored in *shared* variables, so with two Blackhawks in the air, each overwrites the other's data and shutdown removes the wrong listener.

THE CHANGE: both engines initialize, listeners are added once and cleaned up on shutdown, and each helicopter keeps its own data.

YOU'LL TEST: two H-60s running at once; board and exit each repeatedly; engines and electrics stay independent and nothing accumulates.

Laser-code error spamming server logs

BROKEN NOW: Reading the laser channel off the MFD doesn't check that the value is in range, so an out-of-range value crashes the script — a confirmed report with log files showed this error repeating and causing frame drops during a mission.

THE CHANGE: the lookup checks its bounds and falls back to the default code instead of erroring.

YOU'LL TEST: cycle the LST/LRFD channels through everything, fire the laser — log stays clean.

Increase/decrease keybinds do nothing

BROKEN NOW: The keybinds ask the cockpit for a knob spelled "*BALT*", but the knob in the cockpit is spelled "*ALT*" — so the altitude-hold keybind can never find its knob. (A second, smaller fault lives in the shared Hatchet Interaction Framework — a first-use error in the knob-turning code — which we'll report and fix in that project separately.)

THE CHANGE: the name is corrected, and all five modes' increase/decrease binds get verified together.

YOU'LL TEST: every FD increase/decrease keybind visibly changes its mode setting.

PAINT & PARTS TABLE · LATENT

The parts-slot lookup table is off by one

BROKEN NOW: When the ESSS wings were made retexturable, a new entry was inserted into the middle of the master parts-slot list, but the shortcut names that point into that list weren't shifted — five indicator-light entries now point one slot low, the ESSS got no shortcut at all, and the model file lists the same entry in a different position. Nothing currently writes through the broken shortcuts, so players don't see this bug — it's a landmine for the next person who uses them (and reading this table wrong is exactly how we lost days on the console investigation).

THE CHANGE: the shortcuts are realigned with the list, the ESSS gets its entry, and the ordering mismatch is documented.

YOU'LL TEST: nothing visible should change — paint schemes, door numbers, and indicator lights all still work.

CLEANLINESS · 2 BUGS

Debug messages shipping to players

BROKEN NOW: The FLIR auto-scan prints "SETTING UP" chat spam to the player, and the engine code writes a log line every event — both were developer leftovers.

THE CHANGE: both go behind the existing "Show Debug Messages" setting (or away entirely).

YOU'LL TEST: chat and logs stay quiet in normal play.

NAVIGATION · GITHUB #419

First waypoint shows no marker on the TAC map

BROKEN NOW: Creating waypoints in the nav system, the first one in the list never gets its map marker — the classic signature of a list counted from the wrong starting number.

THE CHANGE: confirm the off-by-one in the TAC map page and fix it while we're in that code.

YOU'LL TEST: create several waypoints; every one, including the first, shows on the TAC map.

How it will run

- **One branch, one commit per fix** — each independently revertable, referencing its GitHub issue where one exists (#436, #547, #419).
- **Nothing ships until you've flown it.** Local build first, you test, then the PR — same routine we used for the FLIR and cockpit fixes. The hoist and engine items specifically need a **dedicated-server test**

with two players (or two machines), because that's where those bugs live; everything else is testable solo.

- **No behavior redesigns.** Where a fix touches something players can feel (winch timing), the goal is "same speed as today, just consistent" — anything beyond that waits for its proper phase.

Expected player-facing result: hoist that works for crews on dedicated servers, engines that don't interfere between helicopters, working flight-director keybinds, a working first waypoint marker, and cleaner logs — with zero changes to how anything flies, sounds, or looks.

Technical companion: *H-60 Improvement Plan.md*, Phase 0 table — file paths, line numbers, and the fix for each item. Related but separate: PR #559 (FLIR tracking + copilot cockpit, already done) and issue #560 (View-Gunner LOD for the model team).